

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 04 · Inequalities

With Answers

Questions with full worked solutions

- 19 pages
- Past papers up to June 2026

Inequalities

Your classified route

6

questions

41

marks

2

families

6

variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Inequalities

Families and variants

F01 • One-variable Inequalities

V01 Factorable Quadratic Sign Inequality

V02 Rational Inequality with Critical Points

V03 Discriminant Condition to a Parameter Range

F02 • Graphical Inequality Regions

V01 Shade a Given System of Inequalities

V02 Write a System from a Shaded Region

V03 Derive Boundary Constants, then Write the Region

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One-variable Inequalities

WHAT THIS FAMILY ASKS

Reduce an inequality to factors or a quotient, identify all critical values and use sign or graph logic to state the complete solution set.

VARIANT ROUTE

V01 Factorable Quadratic Sign Inequality

V03 Discriminant Condition to a Parameter Range

2 Complete Classified questions

10 total marks

1.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Solve the inequality

$$4x^2 - 3x + 7 \geq 4x + 9$$

(4)

(Total for Question 1 is 4 marks)

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Question 1

WMA11/01 May/June 2023

T04 Inequalities

F01 One-variable Inequalities

V01 Factorable
Sign Inequality

Quadratic

1

Move all terms to one side $4x^2 - 3x + 7 \geq 4x + 9$ gives $4x^2 - 7x - 2 \geq 0$.

2

Factorise $4x^2 - 7x - 2 = (4x + 1)(x - 2)$.

3

Use the sign of the quadratic

The roots are $x = -\frac{1}{4}$ and $x = 2$. Since the quadratic opens upwards, it is non-negative outside the roots.

FINAL ANSWER $x \leq -\frac{1}{4}$ or $x \geq 2$.

8. The straight line l has equation $y = k(2x - 1)$, where k is a constant.

The curve C has equation $y = x^2 + 2x + 11$

Find the set of values of k for which l does not cross or touch C .

(6)

(Total 6 marks)

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Question 2

WMA11/01 January 2020

T04 Inequalities

F01 One-variable Inequalities

V03 Discriminant Condition
to a Parameter Range

1

Equate the line and curve $k(2x - 1) = x^2 + 2x + 11$, so $x^2 + (2 - 2k)x + (11 + k) = 0$.

2

Use the no-intersection condition

For the line not to cross or touch the curve, the quadratic in x must have no real roots, so its discriminant is less than zero.

3

Form the inequality in k

$(2 - 2k)^2 - 4(1)(11 + k) < 0$. Dividing by 4 gives $(1 - k)^2 - (11 + k) < 0$, so $k^2 - 3k - 10 < 0$.

4

Solve the quadratic inequality

$(k - 5)(k + 2) < 0$, so k lies between the two roots: $-2 < k < 5$.

FINAL ANSWER $-2 < k < 5$.

Graphical Inequality Regions

WHAT THIS FAMILY ASKS

Translate between systems of linear/quadratic inequalities and their coordinate-plane regions, including deriving boundary constants or intersections when necessary.

VARIANT ROUTE

V01 Shade a Given System of Inequalities

V02 Write a System from a Shaded Region

V03 Derive Boundary Constants, then Write the Region

4 Complete Classified questions

31 total marks

3. (i) Find the range of values of z for which

$$\frac{4}{z} < 8$$

giving the answer in simplest form.

(2)

(ii)

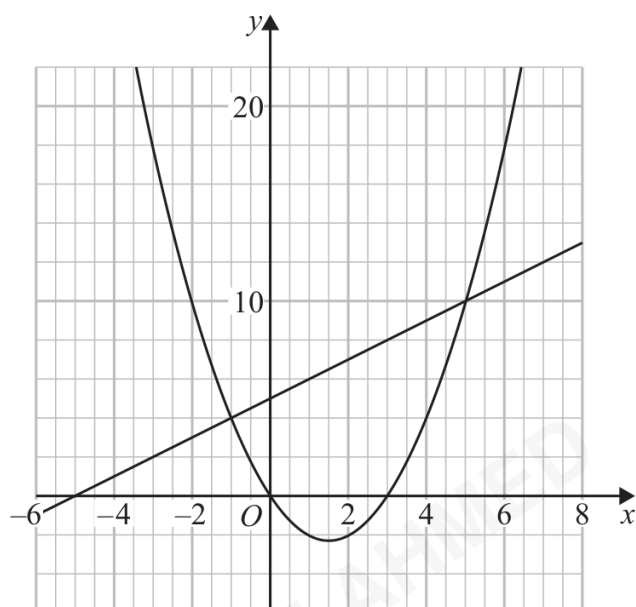


Figure 1

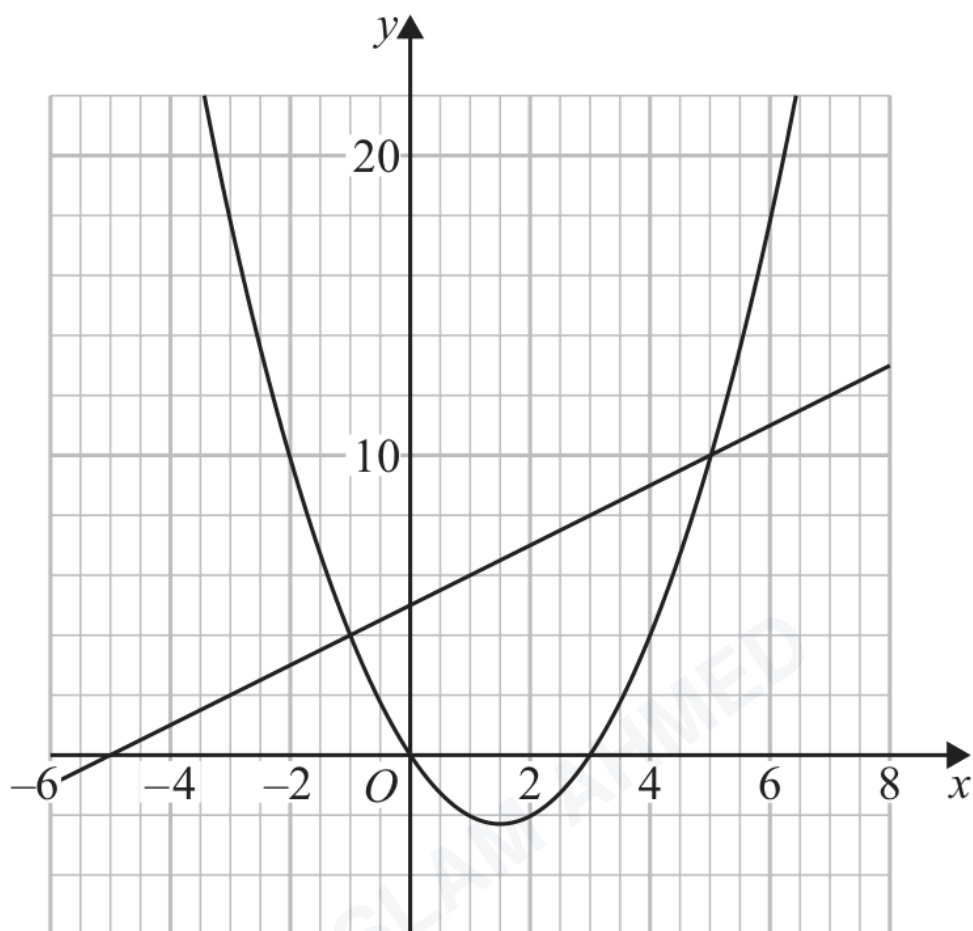
Figure 1 shows a plot of

- the line with equation $y = x + 5$
- the curve with equation $y = x(x - 3)$

On the opposite page there is a copy of Figure 1 labelled Diagram 1.
On Diagram 1 represent the following inequality graphically.

$$x + 5 \leq y \leq x(x - 3) \quad \text{for } x \geq -2$$

(3)

**Diagram 1**

(Total for Question 3 is 5 marks)

Question 3

WMA11/01 January 2026

T04 Inequalities

F02 Graphical Inequality Regions

V01 Shade a Given System of Inequalities

1

Solve the rational inequality $\frac{4}{z} < 8$ gives $\frac{4}{z} - 8 < 0$, so $\frac{4 - 8z}{z} < 0$.

2

Use critical valuesThe critical values are $z = 0$ and $z = \frac{1}{2}$.

3

State the rangeA sign check gives $z < 0$ or $z > \frac{1}{2}$.

4

Find where line and curve meet $x + 5 = x(x - 3)$ gives $x^2 - 4x - 5 = 0$, so $(x - 5)(x + 1) = 0$.

5

Combine with the domainThe inequality $x + 5 \leq y \leq x(x - 3)$ is possible when $x \leq -1$ or $x \geq 5$. With $x \geq -2$, the shaded x-ranges are $-2 \leq x \leq -1$ and $x \geq 5$.

6

Graphical representationDraw the vertical boundary $x = -2$ and shade the two regions between the line and the parabola: from $x = -2$ to $x = -1$, and to the right of $x = 5$.

FINAL ANSWER

(i) $z < 0$ or $z > \frac{1}{2}$. (ii) shade the region above $y = x + 5$, below $y = x(x - 3)$, with $x \geq -2$.

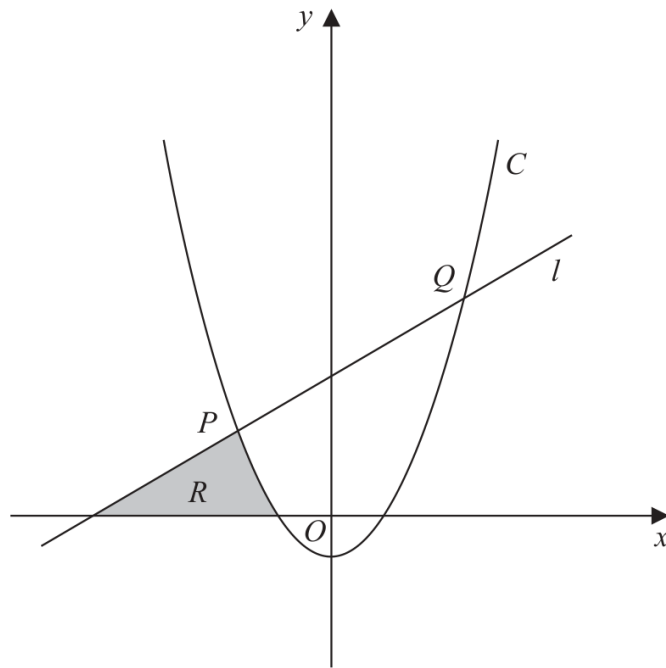


Figure 3

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 3 shows

- the line l with equation $y - 5x = 75$
- the curve C with equation $y = 2x^2 + x - 21$

The line l intersects the curve C at the points P and Q , as shown in Figure 3.

- (a) Find, using algebra, the coordinates of P and the coordinates of Q . (4)

The region R , shown shaded in Figure 3, is bounded by C , l and the x -axis.

- (b) Use inequalities to define the region R . (3)

(Total for Question 6 is 7 marks)

Question 4

WMA11/01 May/June 2024

T04 Inequalities

F02 Graphical Inequality Regions

V02 Write a System from a Shaded Region

1

Write the line as y equals $y - 5x = 75$, so $y = 5x + 75$.

2

Equate line and curve $5x + 75 = 2x^2 + x - 21$, so $2x^2 - 4x - 96 = 0$.

3

Solve the quadratic $x^2 - 2x - 48 = 0 = (x - 8)(x + 6)$, so $x = -6$ or $x = 8$.

4

Find the y -coordinatesUsing $y = 5x + 75$, $x = -6$ gives $y = 45$ and $x = 8$ gives $y = 115$.

5

State P and QFrom the diagram, $P = (-6, 45)$ and $Q = (8, 115)$.

6

Use the x -axis and left branchThe shaded region is above the x -axis and on the left part of the diagram, so $y \geq 0$ and $x \leq 0$.

7

Apply both boundariesThe region is below both the line and the curve, so $y \leq 5x + 75$ and $y \leq 2x^2 + x - 21$.

FINAL ANSWER

(a) $P = (-6, 45)$, $Q = (8, 115)$. (b) $y \geq 0$, $y \leq 5x + 75$, $y \leq 2x^2 + x - 21$, $x \leq 0$.

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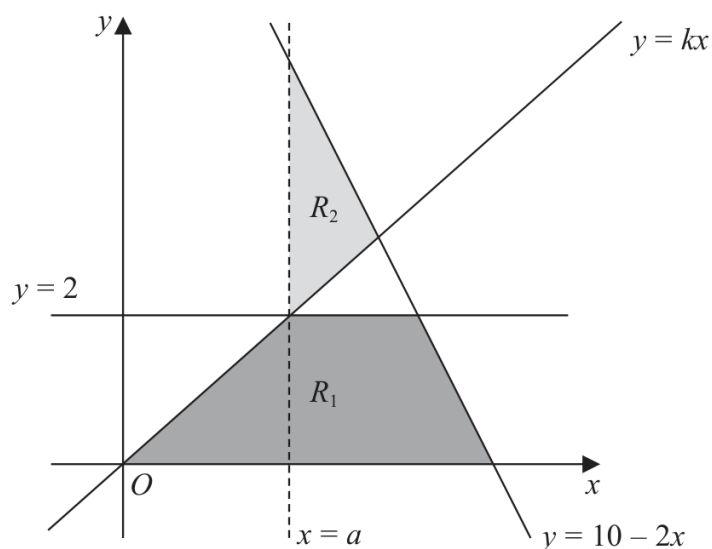


Figure 2

The region R_1 , shown shaded in Figure 2, is defined by the inequalities

$$0 \leq y \leq 2 \quad y \leq 10 - 2x \quad y \leq kx$$

where k is a constant.

The line $x = a$, where a is a constant, passes through the intersection of the lines $y = 2$ and $y = kx$

Given that the area of R_1 is $\frac{27}{4}$ square units,

(a) find

(i) the value of a

(ii) the value of k

(4)

(b) Define the region R_2 , also shown shaded in Figure 2, using inequalities.

(2)

(Total for Question 7 is 6 marks)

Question 5

WMA11/01 May/June 2023

T04 Inequalities

F02 Graphical Inequality Regions

V03 Derive Boundary Constants, then Write the Region

1

Find the area of R1

The bottom horizontal length is 5, from $x = 0$ to the intercept of $y = 10 - 2x$ at $x = 5$.

2

Use the top length

The top horizontal length is $4 - a$, because $y = 2$ meets $y = 10 - 2x$ at $x = 4$.

3

Solve for a

$$\text{Area } R_1 = \frac{1}{2}(5 + 4 - a)(2) = 9 - a = \frac{27}{4}, \text{ so } a = \frac{9}{4}.$$

4

Find k

The line $x = a$ passes through the intersection of $y = 2$ and $y = kx$, so $2 = ka$.

$$\text{Hence } k = \frac{2}{9/4} = \frac{8}{9}.$$

5

Define R2

Region R_2 is to the right of $x = a$, above $y = kx$, and below $y = 10 - 2x$.

FINAL ANSWER

$$(a) a = \frac{9}{4}, k = \frac{8}{9}. \quad (b) x \geq \frac{9}{4}, y \geq \frac{8}{9}x, y \leq 10 - 2x.$$

9.

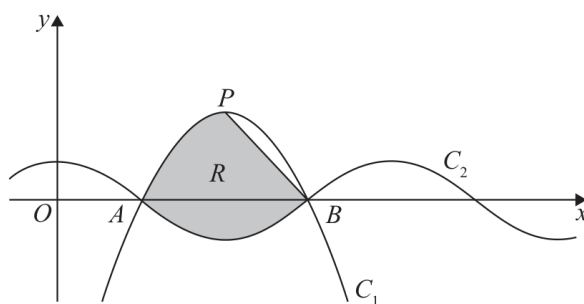


Figure 2

- (a) Express $6x - \frac{27}{4} - x^2$ in the form $a + b(x + c)^2$ where a , b and c are constants to be found.

(3)

Figure 2 shows part of a sketch of curve C_1 with equation

$$y = 6x - \frac{27}{4} - x^2$$

Given that the point P is the maximum point on C_1

- (b) state the coordinates of P

(2)

Figure 2 also shows part of a sketch of curve C_2 with equation

$$y = \cos(kx)$$

where k is a constant and x is measured in radians.

Given that C_1 and C_2 intersect on the x -axis at point A and at point B , as shown in Figure 2,

- (c) (i) state the x coordinate of B
 (ii) state the value of k
 (iii) state the period of C_2

(3)

The line segment L joins P and B .

The region R , shown shaded in Figure 2, is bounded by L , C_1 and C_2

- (d) Use inequalities to define R .

(5)

(Total for Question 9 is 13 marks)

Question 6

WMA11/01 January 2025

T04 Inequalities

F02 Graphical Inequality Regions

V03 Derive Boundary Constants, then Write the Region

1

Complete the square

$$6x - \frac{27}{4} - x^2 = -(x^2 - 6x) - \frac{27}{4} = -(x - 3)^2 + 9 - \frac{27}{4}.$$

2

State the completed-square form

$$6x - \frac{27}{4} - x^2 = \frac{9}{4} - (x - 3)^2.$$

3

Find the maximum pointThe maximum point is $P = (3, \frac{9}{4})$.

4

Find the x-intercepts of C1
 $\frac{9}{4} - (x - 3)^2 = 0$ gives $x = 3 \pm \frac{3}{2}$. Hence A has $x = \frac{3}{2}$ and B has $x = \frac{9}{2}$.

5

Use consecutive zeros of cosineThe distance between consecutive zeros of $\cos(kx)$ is $\frac{\pi}{k}$.

6

Find k and the period
 Since $x_B - x_A = \frac{9}{2} - \frac{3}{2} = 3$, $\frac{\pi}{k} = 3$, so $k = \frac{\pi}{3}$. The period is $\frac{2\pi}{k} = 6$.

7

Find the line through P and B
 The line through $P(3, \frac{9}{4})$ and $B(\frac{9}{2}, 0)$ has gradient $-\frac{3}{2}$, so L is $y = -\frac{3}{2}x + \frac{27}{4}$.

8

Define the x-range

The region lies between A and B , so $\frac{3}{2} \leq x \leq \frac{9}{2}$.

9

Define the lower boundary

The lower boundary is C_2 , so $y \geq \cos\left(\frac{\pi}{3}x\right)$.

10

Define the upper boundaries

The region is below both C_1 and the line segment L , so $y \leq \frac{9}{4} - (x-3)^2$ and $y \leq -\frac{3}{2}x + \frac{27}{4}$.

FINAL ANSWER

(a) $\frac{9}{4} - (x-3)^2$. (b) $P = (3, \frac{9}{4})$. (c)(i) $x_B = \frac{9}{2}$. (c)(ii) $k = \frac{\pi}{3}$.
 (c)(iii) 6. (d) $\frac{3}{2} \leq x \leq \frac{9}{2}$, $y \geq \cos(\frac{\pi}{3}x)$, $y \leq \frac{9}{4} - (x-3)^2$, $y \leq -\frac{3}{2}x + \frac{27}{4}$.